```
<!DOCTYPE html>
<html lang="pt-BR">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>QMFXperience - Laboratório Virtual de Química Medicinal e Farmacêutica</title>

  <!-- Tailwind CSS -->
  <script src="https://cdn.tailwindcss.com"></script>

  <!-- FontAwesome -->
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">

  <!-- Chart.js -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

  <style>
    @import
url('https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800&family=
JetBrains+Mono:wght@400;500;700&display=swap');

    body {
      font-family: 'Inter', sans-serif;
      background-color: #0f172a;
      color: #f8fafc;
    }

    .code-font {
      font-family: 'JetBrains Mono', monospace;
    }

    .glass-card {
      background: rgba(30, 41, 59, 0.7);
      backdrop-filter: blur(12px);
      border: 1px solid rgba(255, 255, 255, 0.1);
    }

    .glass-nav {
      background: rgba(15, 23, 42, 0.85);
      backdrop-filter: blur(16px);
      border-bottom: 1px solid rgba(255, 255, 255, 0.1);
    }

    /* Custom Liquid Animation */
    @keyframes wave {
      0%, 100% { transform: translateY(0) rotate(0deg); }
      50% { transform: translateY(-3px) rotate(1deg); }
```

```
    }

    .liquid-wave {
      animation: wave 3s infinite ease-in-out;
    }

    /* TLC Plate Canvas */
    .tlc-plate {
      background: #e2e8f0;
      border-radius: 4px;
      box-shadow: inset 0 0 10px rgba(0,0,0,0.2);
    }

    /* Custom scrollbar */
    ::-webkit-scrollbar {
      width: 8px;
      height: 8px;
    }
    ::-webkit-scrollbar-track {
      background: #0f172a;
    }
    ::-webkit-scrollbar-thumb {
      background: #334155;
      border-radius: 4px;
    }
    ::-webkit-scrollbar-thumb:hover {
      background: #475569;
    }
  </style>
</head>
<body class="min-h-screen flex flex-col justify-between">

  <!-- HEADER / NAVBAR -->
  <header class="sticky top-0 z-50 glass-nav px-4 lg:px-8 py-3 flex items-center
justify-between">
    <div class="flex items-center space-x-3 cursor-pointer" onclick="showModule('home')">
      <div class="w-10 h-10 rounded-xl bg-gradient-to-tr from-cyan-500 to-blue-600 flex
items-center justify-center text-white font-black text-xl shadow-lg shadow-cyan-500/20">
        Q
      </div>
      <div>
        <h1 class="font-bold text-lg leading-tight tracking-wide bg-clip-text text-transparent
bg-gradient-to-r from-cyan-400 via-sky-300 to-blue-500">
          QMFXperience
        </h1>
        <p class="text-xs text-slate-400 font-medium">UFPR • Química Medicinal e
Farmacêutica</p>
      </div>
```

```
    </div>

    <!-- Navigation Pills -->
    <nav class="hidden md:flex items-center space-x-1 bg-slate-800/80 p-1 rounded-xl border
border-slate-700/50">
      <button onclick="showModule('p1')" id="nav-p1" class="px-3 py-1.5 text-xs font-semibold
rounded-lg transition-all text-slate-300 hover:text-white hover:bg-slate-700/50">
        1. Sinvastatina (TLC)
      </button>
      <button onclick="showModule('p2')" id="nav-p2" class="px-3 py-1.5 text-xs font-semibold
rounded-lg transition-all text-slate-300 hover:text-white hover:bg-slate-700/50">
        2. Ácido Mandélico (log P)
      </button>
      <button onclick="showModule('p3')" id="nav-p3" class="px-3 py-1.5 text-xs font-semibold
rounded-lg transition-all text-slate-300 hover:text-white hover:bg-slate-700/50">
        3. Captopril (pKa)
      </button>
      <button onclick="showModule('p4')" id="nav-p4" class="px-3 py-1.5 text-xs font-semibold
rounded-lg transition-all text-slate-300 hover:text-white hover:bg-slate-700/50">
        4. Cetoconazol (UV/Ion)
      </button>
      <button onclick="showModule('p5')" id="nav-p5" class="px-3 py-1.5 text-xs font-semibold
rounded-lg transition-all text-slate-300 hover:text-white hover:bg-slate-700/50">
        5. Sulfonamidas (QSAR)
      </button>
    </nav>

    <!-- Quiz & Profile -->
    <div class="flex items-center space-x-3">
      <button onclick="showModule('quiz')" id="nav-quiz" class="flex items-center space-x-2
bg-gradient-to-r from-amber-500 to-orange-600 hover:from-amber-600 hover:to-orange-700
text-white px-3 py-1.5 rounded-lg text-xs font-bold transition-all shadow-md
shadow-orange-500/20">
        <i class="fa-solid fa-trophy"></i>
        <span>Quiz & Gamificação</span>
      </button>
    </div>
  </header>

  <!-- MAIN CONTENT CONTAINER -->
  <main class="flex-grow container mx-auto px-4 py-6">

    <!-- HOME / WELCOME MODULE -->
    <section id="mod-home" class="space-y-8">
      <div class="glass-card rounded-2xl p-8 lg:p-12 relative overflow-hidden border
border-cyan-500/20 shadow-2xl">
        <div class="absolute -right-20 -bottom-20 w-80 h-80 bg-cyan-500/10 rounded-full
blur-3xl pointer-events-none"></div>
```

```
      <div class="absolute -left-20 -top-20 w-80 h-80 bg-blue-600/10 rounded-full blur-3xl pointer-events-none"></div>

      <div class="max-w-3xl space-y-4 relative z-10">
        <span class="inline-flex items-center space-x-2 bg-cyan-500/10 border border-cyan-500/30 text-cyan-400 px-3 py-1 rounded-full text-xs font-semibold tracking-wider uppercase">
          <i class="fa-solid fa-flask"></i>
          <span>Plataforma Educacional Interativa</span>
        </span>
        <h2 class="text-3xl lg:text-5xl font-extrabold tracking-tight text-white leading-tight">
          Laboratório Virtual de <br>
          <span class="bg-clip-text text-transparent bg-gradient-to-r from-cyan-400 via-sky-300 to-blue-500">
            Química Medicinal e Farmacêutica
          </span>
        </h2>
        <p class="text-slate-300 text-sm lg:text-base leading-relaxed">
          Bem-vindo ao portal interativo da disciplina de QMF da UFPR. Explore experimentos práticos simulados, conceitos de físico-química farmacêutica, ionização, coeficientes de partição ($\log P$), espectrofotometria UV, cromatografia em camada delgada (CCD) e Relações Quantitativas Estrutura-Atividade (QSAR).
        </p>
      </div>
    </div>

    <!-- Modules Grid -->
    <div class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">

      <!-- Card 1 -->
      <div onclick="showModule('p1')" class="glass-card p-6 rounded-xl hover:border-cyan-500/50 hover:bg-slate-800/80 transition-all cursor-pointer group flex flex-col justify-between">
        <div>
          <div class="w-12 h-12 rounded-lg bg-cyan-500/10 border border-cyan-500/20 flex items-center justify-center text-cyan-400 text-xl mb-4 group-hover:scale-110 transition-transform">
            <i class="fa-solid fa-vial"></i>
          </div>
          <h3 class="text-lg font-bold text-white mb-2 group-hover:text-cyan-400 transition-colors">1. Sinvastatina & Pró-fármacos</h3>
          <p class="text-xs text-slate-400 leading-relaxed">
            Bioativação por hidrólise alcalina da lactona e análise de purificação por Cromatografia em Camada Delgada (CCD-F254).
          </p>
        </div>
        <div class="mt-6 flex items-center text-xs font-semibold text-cyan-400 space-x-1">
          <span>Iniciar Prática</span>
```

```
          <i class="fa-solid fa-arrow-right text-[10px] group-hover:translate-x-1 transition-transform"></i>
        </div>
      </div>

      <!-- Card 2 -->
      <div onclick="showModule('p2')" class="glass-card p-6 rounded-xl hover:border-cyan-500/50 hover:bg-slate-800/80 transition-all cursor-pointer group flex flex-col justify-between">
        <div>
          <div class="w-12 h-12 rounded-lg bg-blue-500/10 border border-blue-500/20 flex items-center justify-center text-blue-400 text-xl mb-4 group-hover:scale-110 transition-transform">
            <i class="fa-solid fa-filter"></i>
          </div>
          <h3 class="text-lg font-bold text-white mb-2 group-hover:text-blue-400 transition-colors">2. Ácido Mandélico & Lipofilicidade</h3>
          <p class="text-xs text-slate-400 leading-relaxed">
            Determinação experimental do Coeficiente de Partição ($\log P$) por extração líquido-líquido em Funil de Separação.
          </p>
        </div>
        <div class="mt-6 flex items-center text-xs font-semibold text-blue-400 space-x-1">
          <span>Iniciar Prática</span>
          <i class="fa-solid fa-arrow-right text-[10px] group-hover:translate-x-1 transition-transform"></i>
        </div>
      </div>

      <!-- Card 3 -->
      <div onclick="showModule('p3')" class="glass-card p-6 rounded-xl hover:border-cyan-500/50 hover:bg-slate-800/80 transition-all cursor-pointer group flex flex-col justify-between">
        <div>
          <div class="w-12 h-12 rounded-lg bg-indigo-500/10 border border-indigo-500/20 flex items-center justify-center text-indigo-400 text-xl mb-4 group-hover:scale-110 transition-transform">
            <i class="fa-solid fa-chart-line"></i>
          </div>
          <h3 class="text-lg font-bold text-white mb-2 group-hover:text-indigo-400 transition-colors">3. Captopril & Ionização ($pK_a$)</h3>
          <p class="text-xs text-slate-400 leading-relaxed">
            Titulação potenciométrica com pHmetro, cálculo do ponto de equivalência com 1ª e 2ª derivadas e Henderson-Hasselbalch.
          </p>
        </div>
        <div class="mt-6 flex items-center text-xs font-semibold text-indigo-400 space-x-1">
          <span>Iniciar Prática</span>
```

```
          <i class="fa-solid fa-arrow-right text-[10px] group-hover:translate-x-1 transition-transform"></i>
        </div>
      </div>

      <!-- Card 4 -->
      <div onclick="showModule('p4')" class="glass-card p-6 rounded-xl hover:border-cyan-500/50 hover:bg-slate-800/80 transition-all cursor-pointer group flex flex-col justify-between">
        <div>
          <div class="w-12 h-12 rounded-lg bg-sky-500/10 border border-sky-500/20 flex items-center justify-center text-sky-400 text-xl mb-4 group-hover:scale-110 transition-transform">
            <i class="fa-solid fa-wave-square"></i>
          </div>
          <h3 class="text-lg font-bold text-white mb-2 group-hover:text-sky-400 transition-colors">4. Cetoconazol & Absorção UV</h3>
          <p class="text-xs text-slate-400 leading-relaxed">
            Dissolução pH-dependente e análise espectrofotométrica. Efeito de antiácidos e ion trapping na secreção láctea.
          </p>
        </div>
        <div class="mt-6 flex items-center text-xs font-semibold text-sky-400 space-x-1">
          <span>Iniciar Prática</span>
          <i class="fa-solid fa-arrow-right text-[10px] group-hover:translate-x-1 transition-transform"></i>
        </div>
      </div>

      <!-- Card 5 -->
      <div onclick="showModule('p5')" class="glass-card p-6 rounded-xl hover:border-cyan-500/50 hover:bg-slate-800/80 transition-all cursor-pointer group flex flex-col justify-between">
        <div>
          <div class="w-12 h-12 rounded-lg bg-emerald-500/10 border border-emerald-500/20 flex items-center justify-center text-emerald-400 text-xl mb-4 group-hover:scale-110 transition-transform">
            <i class="fa-solid fa-dna"></i>
          </div>
          <h3 class="text-lg font-bold text-white mb-2 group-hover:text-emerald-400 transition-colors">5. Sulfonamidas & QSAR</h3>
          <p class="text-xs text-slate-400 leading-relaxed">
            CCD em Fase Reversa (C18), constante substituinte de Hansch ($\pi$), $R_m$ e correlação parabólica com a atividade antibacteriana.
          </p>
        </div>
        <div class="mt-6 flex items-center text-xs font-semibold text-emerald-400 space-x-1">
          <span>Iniciar Prática</span>
```

```
          <i class="fa-solid fa-arrow-right text-[10px] group-hover:translate-x-1
transition-transform"></i>
        </div>
      </div>

      <!-- Card Quiz -->
      <div onclick="showModule('quiz')" class="glass-card p-6 rounded-xl border
border-amber-500/30 hover:border-amber-500/60 bg-gradient-to-br from-amber-500/5
to-orange-500/10 hover:bg-slate-800/80 transition-all cursor-pointer group flex flex-col
justify-between">
        <div>
          <div class="w-12 h-12 rounded-lg bg-amber-500/10 border border-amber-500/20 flex
items-center justify-center text-amber-400 text-xl mb-4 group-hover:scale-110
transition-transform">
            <i class="fa-solid fa-award"></i>
          </div>
          <h3 class="text-lg font-bold text-amber-300 mb-2">Desafio & Gamificação</h3>
          <p class="text-xs text-slate-400 leading-relaxed">
            Teste seus conhecimentos com problemas reais da feira, receba pontuação
instantânea e atinja a pontuação máxima.
          </p>
        </div>
        <div class="mt-6 flex items-center text-xs font-semibold text-amber-400 space-x-1">
          <span>Acessar Quiz</span>
          <i class="fa-solid fa-arrow-right text-[10px] group-hover:translate-x-1
transition-transform"></i>
        </div>
      </div>

    </div>
  </section>

  <!-- PRÁTICA 1: SINVASTATINA -->
  <section id="mod-p1" class="hidden space-y-6">
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b
border-slate-800 pb-4">
      <div>
        <span class="text-xs font-bold text-cyan-400 uppercase tracking-widest">Prática 01 •
Síntese & Purificação</span>
        <h2 class="text-2xl font-bold text-white">Bioativação da Sinvastatina e Análise em
CCD</h2>
      </div>
      <div class="flex space-x-2">
        <button onclick="showModule('home')" class="px-3 py-1.5 rounded-lg bg-slate-800
hover:bg-slate-700 text-xs font-semibold text-slate-300">
          <i class="fa-solid fa-house mr-1"></i> Início
        </button>
      </div>
```

```
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-12 gap-6">
      <!-- Interactive Lab Area -->
      <div class="lg:col-col-span-12 xl:col-span-7 glass-card p-6 rounded-xl space-y-6">
        <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center
space-x-2">
          <i class="fa-solid fa-flask-vial text-cyan-400"></i>
          <span>Simulador do Processo de Reação e CCD</span>
        </h3>

        <!-- Control Steps -->
        <div class="grid grid-cols-3 gap-2">
          <button onclick="setP1Step(1)" id="p1-btn-1" class="p-3 rounded-lg border text-left
transition-all bg-cyan-500/20 border-cyan-500 text-white">
            <div class="text-[10px] text-cyan-300 font-mono">PASSO 1</div>
            <div class="text-xs font-bold">Hidrólise Alcalina</div>
          </button>
          <button onclick="setP1Step(2)" id="p1-btn-2" class="p-3 rounded-lg border
border-slate-700 bg-slate-800/50 text-slate-400 hover:bg-slate-700/50">
            <div class="text-[10px] font-mono">PASSO 2</div>
            <div class="text-xs font-bold">Aplicação na Placa</div>
          </button>
          <button onclick="setP1Step(3)" id="p1-btn-3" class="p-3 rounded-lg border
border-slate-700 bg-slate-800/50 text-slate-400 hover:bg-slate-700/50">
            <div class="text-[10px] font-mono">PASSO 3</div>
            <div class="text-xs font-bold">Revelação UV / $R_f$</div>
          </button>
        </div>

        <!-- Dynamic Display Box -->
        <div class="bg-slate-900/80 rounded-xl p-6 border border-slate-800 min-h-[300px] flex
flex-col justify-center items-center relative">

          <!-- Step 1 View: Chemical Reaction -->
          <div id="p1-view-1" class="w-full text-center space-y-4">
            <div class="flex items-center justify-center space-x-6">
              <!-- Sinvastatina Lactona -->
              <div class="p-4 bg-slate-800/80 rounded-lg border border-slate-700">
                <div class="w-24 h-24 mx-auto flex items-center justify-center text-slate-400">
                  <i class="fa-solid fa-ring text-4xl text-cyan-400"></i>
                </div>
                <span class="text-xs font-bold text-slate-200 block mt-2">Sinvastatina</span>
                <span class="text-[10px] text-cyan-400 font-mono block">Lactona
(Pró-fármaco)</span>
                <span class="text-[10px] text-slate-400 block mt-1">Apolar / Lipofílica</span>
              </div>
```

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            <div class="text-slate-500 font-bold text-lg">
              <i class="fa-solid fa-plus block text-xs"></i>
              <span class="text-xs font-mono text-cyan-400">NaOH/EtOH</span>
              <i class="fa-solid fa-right-long text-xl block text-cyan-400"></i>
            </div>

            <!-- Ácido Sinvastatínico -->
            <div class="p-4 bg-slate-800/80 rounded-lg border border-slate-700">
              <div class="w-24 h-24 mx-auto flex items-center justify-center text-slate-400">
                <i class="fa-solid fa-vial-circle-check text-4xl text-emerald-400"></i>
              </div>
              <span class="text-xs font-bold text-slate-200 block mt-2">Ácido
Sinvastatínico</span>
              <span class="text-[10px] text-emerald-400 font-mono block">Forma Ativa
($COO^-$)</span>
              <span class="text-[10px] text-slate-400 block mt-1">Polar / Ionizada</span>
            </div>
          </div>
          <p class="text-xs text-slate-400 max-w-md mx-auto">
            A abertura do anel lactônico por saponificação expõe o grupo carboxilato,
transformando o pró-fármaco inativo na sua forma ativa inibidora da HMG-CoA redutase.
          </p>
        </div>

        <!-- Step 2 & 3 View: TLC Plate -->
        <div id="p1-view-2" class="hidden w-full flex flex-col items-center space-y-4">
          <div class="flex items-center space-x-8">
            <!-- TLC Chamber / Plate -->
            <div class="relative w-48 h-64 bg-slate-200 rounded-lg p-3 shadow-inner flex
flex-col justify-between border-2 border-slate-400">
              <!-- Solvent Front Line -->
              <div class="w-full border-b border-dashed border-slate-500 flex justify-between
items-center text-[9px] text-slate-600 font-mono">
                <span>Frente do Eluente</span>
                <span>8.0 cm</span>
              </div>

              <!-- Spots Container -->
              <div class="relative flex-grow w-full my-2">
                <!-- Spot 1: Sinvastatina (Apolar - Sobe mais) -->
                <div id="spot-sinva" class="absolute left-1/4 transform -translate-x-1/2 w-6 h-6
rounded-full bg-cyan-600/80 border border-cyan-800 flex items-center justify-center
text-[9px] text-white font-bold transition-all duration-1000" style="bottom: 70%;">
                  S
                </div>

                <!-- Spot 2: Ácido Sinvastatínico (Polar - Fica retido) -->
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```
                <div id="spot-acid" class="absolute left-3/4 transform -translate-x-1/2 w-6 h-6
rounded-full bg-emerald-600/80 border border-emerald-800 flex items-center justify-center
text-[9px] text-white font-bold transition-all duration-1000" style="bottom: 25%;">
                  A
                </div>
              </div>

              <!-- Origin Line -->
              <div class="w-full border-t border-dashed border-slate-500 flex justify-between
items-center text-[9px] text-slate-600 font-mono">
                <span>Linha de Aplicação</span>
                <span>0.0 cm</span>
              </div>
            </div>

            <!-- Legend & Values -->
            <div class="space-y-4 text-xs">
              <div class="p-3 bg-slate-800 rounded-lg border border-slate-700">
                <span class="font-bold text-cyan-400 block mb-1">Amostra (S):
Sinvastatina</span>
                <p class="text-slate-400 text-[11px]">Distância percorrida: <span
class="text-white font-mono">5.6 cm</span></p>
                <p class="text-slate-400 text-[11px]">$R_f = 5.6 / 8.0 = $ <span
class="text-cyan-400 font-bold font-mono">0.70</span></p>
              </div>

              <div class="p-3 bg-slate-800 rounded-lg border border-slate-700">
                <span class="font-bold text-emerald-400 block mb-1">Amostra (A): Ácido
Sinvastatínico</span>
                <p class="text-slate-400 text-[11px]">Distância percorrida: <span
class="text-white font-mono">2.0 cm</span></p>
                <p class="text-slate-400 text-[11px]">$R_f = 2.0 / 8.0 = $ <span
class="text-emerald-400 font-bold font-mono">0.25</span></p>
              </div>
            </div>
          </div>
        </div>

      </div>
    </div>

    <!-- Scientific Discussion & ADME -->
    <div class="lg:col-span-12 xl:col-span-5 glass-card p-6 rounded-xl space-y-4">
      <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center
space-x-2">
        <i class="fa-solid fa-microscope text-cyan-400"></i>
        <span>Fundamentação Científica & QSAR</span>
      </h3>
```

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        <div class="space-y-3 text-xs leading-relaxed text-slate-300">
          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700">
            <h4 class="font-bold text-cyan-300 mb-1">Por que a Sinvastatina é um Pró-fármaco?</h4>
            <p class="text-slate-400">
              A lactona fecha a estrutura lipofílica, permitindo uma absorção intestinal eficiente por difusão passiva. No fígado, é hidrolisada pelas esterases hepáticas para formar o hydroxyacid ativo.
            </p>
          </div>

          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700">
            <h4 class="font-bold text-sky-300 mb-1">Interpretação da CCD em Fase Normal (Sílica Gel):</h4>
            <p class="text-slate-400">
              A sílica gel é altamento polar. Substâncias polares (como o ácido ativado ionizado) interagem fortemente com a fase estacionária, resultando em menor retenção na fase móvel e menor $R_f$ ($R_f = 0.25$). A lactona lipofílica elui mais rapidamente ($R_f = 0.70$).
            </p>
          </div>
        </div>
      </div>
    </div>
  </section>

  <!-- PRÁTICA 2: ÁCIDO MANDÉLICO -->
  <section id="mod-p2" class="hidden space-y-6">
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-800 pb-4">
      <div>
        <span class="text-xs font-bold text-blue-400 uppercase tracking-widest">Prática 02 • Físico-Química Farmacêutica</span>
        <h2 class="text-2xl font-bold text-white">Determinação do Coeficiente de Partição ($\log P$) do Ácido Mandélico</h2>
      </div>
      <button onclick="showModule('home')" class="px-3 py-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-xs font-semibold text-slate-300">
        <i class="fa-solid fa-house mr-1"></i> Início
      </button>
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-12 gap-6">
      <!-- Interactive Funnel Simulator -->
      <div class="lg:col-span-12 xl:col-span-7 glass-card p-6 rounded-xl space-y-6">
        <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center space-x-2">
```

```
          <i class="fa-solid fa-filter text-blue-400"></i>
          <span>Simulador de Extração Líquido-Líquido em Funil de Separação</span>
        </h3>

        <div class="grid grid-cols-1 md:grid-cols-2 gap-6 items-center">
          <!-- Funnel Graphic -->
          <div class="bg-slate-900/90 rounded-xl p-6 border border-slate-800 flex flex-col
items-center justify-center min-h-[320px] relative">
            <div class="w-32 h-56 relative flex flex-col items-center">
              <!-- Top Cap -->
              <div class="w-8 h-4 bg-slate-600 rounded-t-md mb-1"></div>
              <!-- Funnel Body -->
              <div class="w-28 h-36 bg-slate-800/80 border-2 border-slate-600 rounded-b-full
overflow-hidden relative flex flex-col justify-end">
                <!-- Organic Phase (Eter Etilico - Top) -->
                <div id="phase-org" class="w-full bg-amber-500/30 border-b
border-amber-400/50 flex items-center justify-center transition-all duration-500"
style="height: 50%;">
                  <span class="text-[10px] font-bold text-amber-300">Fase Orgânica
(Éter)</span>
                </div>
                <!-- Aqueous Phase (Water - Bottom) -->
                <div id="phase-aq" class="w-full bg-cyan-600/40 flex items-center justify-center
transition-all duration-500" style="height: 50%;">
                  <span class="text-[10px] font-bold text-cyan-200">Fase Aquosa
($H_2O$)</span>
                </div>
              </div>
              <!-- Tap Valve -->
              <div class="w-6 h-6 bg-slate-700 border border-slate-500 my-1
rounded-sm"></div>
              <div class="w-2 h-8 bg-slate-600 rounded-b-md"></div>
            </div>

            <button onclick="runExtraction()" class="mt-4 px-4 py-2 bg-blue-600
hover:bg-blue-500 text-white font-bold text-xs rounded-lg shadow-lg transition-all flex
items-center space-x-2">
              <i class="fa-solid fa-arrows-rotate"></i>
              <span>Agitar e Equilibrar Fases</span>
            </button>
          </div>

          <!-- Controls and Titration Result -->
          <div class="space-y-4">
            <div class="bg-slate-800/60 p-4 rounded-xl border border-slate-700 space-y-3">
              <label class="text-xs font-bold text-slate-300 block">Volume do Solvente Orgânico
(Éter Etílico):</label>
              <div class="flex items-center space-x-3">
```

```
            <input type="range" id="vol-eter" min="10" max="50" value="25" class="w-full
accent-blue-500" oninput="updateExtraction()">
            <span id="vol-eter-val" class="text-xs font-mono font-bold text-blue-400">25
mL</span>
          </div>

          <label class="text-xs font-bold text-slate-300 block mt-2">pH do Meio
Aquoso:</label>
          <select id="p2-ph" onchange="updateExtraction()" class="w-full bg-slate-900
border border-slate-700 text-xs rounded-lg p-2 text-slate-200">
            <option value="2.0">pH 2.0 (Totalmente Não-Ionizado)</option>
            <option value="3.8">pH 3.8 (pH = pKa, 50% Ionizado)</option>
            <option value="6.0">pH 6.0 (Totalmente Ionizado)</option>
          </select>
        </div>

        <!-- Output Box -->
        <div class="bg-slate-900 p-4 rounded-xl border border-blue-500/30 space-y-2">
          <div class="flex justify-between items-center text-xs">
            <span class="text-slate-400">Massa Inicial:</span>
            <span class="font-mono text-white font-bold">1.00 g</span>
          </div>
          <div class="flex justify-between items-center text-xs">
            <span class="text-slate-400">Concentração Fase Orgânica ($C_{org}$):</span>
            <span id="c-org-val" class="font-mono text-amber-400 font-bold">0.024
M</span>
          </div>
          <div class="flex justify-between items-center text-xs">
            <span class="text-slate-400">Concentração Fase Aquosa ($C_{aq}$):</span>
            <span id="c-aq-val" class="font-mono text-cyan-400 font-bold">0.008 M</span>
          </div>
          <div class="pt-2 border-t border-slate-800 flex justify-between items-center">
            <span class="text-xs font-bold text-slate-200">Log P Aparente / Medido:</span>
            <span id="logp-val" class="text-sm font-black font-mono
text-blue-400">0.48</span>
          </div>
        </div>
      </div>
    </div>
  </div>

  <!-- Discussion Section -->
  <div class="lg:col-span-12 xl:col-span-5 glass-card p-6 rounded-xl space-y-4">
    <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center
space-x-2">
      <i class="fa-solid fa-book-bookmark text-blue-400"></i>
      <span>Equação & Conceito Teórico</span>
    </h3>
```

```
        <div class="space-y-3 text-xs leading-relaxed text-slate-300">
          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700">
            <h4 class="font-bold text-blue-300 mb-1">Coeficiente de Partição ($P$):</h4>
            <p class="text-slate-400 mb-2">
              O coeficiente de partição mede a razão de solubilidade do fármaco não ionizado entre uma fase orgânica não polar (ex: octanol ou éter) e uma fase aquosa:
            </p>
            <div class="bg-slate-900 p-2 rounded text-center code-font text-blue-400 font-bold">
              P = [C_orgânica] / [C_aquosa]
            </div>
          </div>

          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700">
            <h4 class="font-bold text-blue-300 mb-1">Influência da Ionização ($pK_a = 3.85$):</h4>
            <p class="text-slate-400">
              Conforme o pH da fase aquosa aumenta acima do $pK_a$, o grupo carboxila ($-COOH$) desprotona para formato ($-COO^-$). Espécies ionizadas são extremamente solúveis em água, reduzindo drasticamente a partição para a fase orgânica ($\log D$ diminui).
            </p>
          </div>
        </div>
      </div>
    </div>
  </section>

  <!-- PRÁTICA 3: CAPTOPRIL -->
  <section id="mod-p3" class="hidden space-y-6">
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-800 pb-4">
      <div>
        <span class="text-xs font-bold text-indigo-400 uppercase tracking-widest">Prática 03 • Controle de Qualidade & Potenciometria</span>
        <h2 class="text-2xl font-bold text-white">Determinação Potenciométrica do $pK_a$ do Captopril</h2>
      </div>
      <button onclick="showModule('home')" class="px-3 py-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-xs font-semibold text-slate-300">
        <i class="fa-solid fa-house mr-1"></i> Início
      </button>
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-12 gap-6">
      <!-- Interactive Chart & Titration -->
      <div class="lg:col-span-12 xl:col-span-7 glass-card p-6 rounded-xl space-y-4">
```

```
        <div class="flex justify-between items-center">
          <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex
items-center space-x-2">
            <i class="fa-solid fa-chart-line text-indigo-400"></i>
            <span>Curva de Titulação Potenciométrica (Captopril vs. NaOH 0.1M)</span>
          </h3>

          <!-- Toggle Derivative Buttons -->
          <div class="flex space-x-1 text-[10px]">
            <button onclick="updateP3Chart('raw')" id="btn-p3-raw" class="px-2 py-1 rounded
bg-indigo-600 text-white font-bold">Curva pH</button>
            <button onclick="updateP3Chart('d1')" id="btn-p3-d1" class="px-2 py-1 rounded
bg-slate-800 text-slate-400 hover:text-white">1ª Derivada</button>
            <button onclick="updateP3Chart('d2')" id="btn-p3-d2" class="px-2 py-1 rounded
bg-slate-800 text-slate-400 hover:text-white">2ª Derivada</button>
          </div>
        </div>

        <!-- Chart Container -->
        <div class="h-64 bg-slate-900/80 p-2 rounded-xl border border-slate-800">
          <canvas id="captoprilChart"></canvas>
        </div>

        <!-- Interactive Values Card -->
        <div class="grid grid-cols-3 gap-3">
          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700 text-center">
            <span class="text-[10px] text-slate-400 block font-semibold">1º $pK_a$
(Carboxila)</span>
            <span class="text-base font-bold font-mono text-indigo-400">3.70</span>
          </div>
          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700 text-center">
            <span class="text-[10px] text-slate-400 block font-semibold">2º $pK_a$ (Tiol
-SH)</span>
            <span class="text-base font-bold font-mono text-cyan-400">9.80</span>
          </div>
          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700 text-center">
            <span class="text-[10px] text-slate-400 block font-semibold">Ponto
Equivalência</span>
            <span class="text-base font-bold font-mono text-emerald-400">12.5 mL</span>
          </div>
        </div>
      </div>

      <!-- Henderson-Hasselbalch Calculator -->
      <div class="lg:col-span-12 xl:col-span-5 glass-card p-6 rounded-xl space-y-4">
        <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center
space-x-2">
          <i class="fa-solid fa-calculator text-indigo-400"></i>
```

```
          <span>Simulador de Ionização (Henderson-Hasselbalch)</span>
        </h3>

        <div class="space-y-4">
          <div class="bg-slate-800/60 p-4 rounded-xl border border-slate-700 space-y-3">
            <label class="text-xs font-bold text-slate-300 block">Selecione o pH Fisiológico do
Meio:</label>
            <input type="range" id="p3-ph-slider" min="1.0" max="10.0" step="0.1" value="2.0"
class="w-full accent-indigo-500" oninput="calcIonization()">
            <div class="flex justify-between items-center text-xs font-mono">
              <span class="text-slate-400">pH Selecionado:</span>
              <span id="p3-ph-val" class="font-bold text-indigo-400 text-sm">2.0</span>
            </div>
          </div>

          <!-- Ionization Results -->
          <div class="bg-slate-900 p-4 rounded-xl border border-indigo-500/30 space-y-3">
            <span class="text-xs font-bold text-slate-300 block">Grau de Ionização do
Captopril:</span>

            <!-- Carboxyl Progress -->
            <div>
              <div class="flex justify-between text-[11px] mb-1">
                <span class="text-slate-400">Carboxila ($-COOH \rightarrow -COO^-$)</span>
                <span id="p3-carboxyl-val" class="font-mono text-indigo-400
font-bold">1.96%</span>
              </div>
              <div class="w-full bg-slate-800 rounded-full h-2">
                <div id="p3-carboxyl-bar" class="bg-indigo-500 h-2 rounded-full transition-all
duration-300" style="width: 2%;"></div>
              </div>
            </div>

            <!-- Thiol Progress -->
            <div>
              <div class="flex justify-between text-[11px] mb-1">
                <span class="text-slate-400">Tiol ($-SH \rightarrow -S^-$)</span>
                <span id="p3-thiol-val" class="font-mono text-cyan-400 font-bold">0.00%</span>
              </div>
              <div class="w-full bg-slate-800 rounded-full h-2">
                <div id="p3-thiol-bar" class="bg-cyan-500 h-2 rounded-full transition-all
duration-300" style="width: 0%;"></div>
              </div>
            </div>

            <div class="pt-2 border-t border-slate-800 text-[11px] text-slate-400">
              <span class="font-bold text-slate-200">Local de Maior Absorção:</span>
```

```
            <span id="p3-site-val" class="block font-semibold text-emerald-400
mt-0.5">Estômago (pH 1.5 - 2.0) - Predomina não-ionizado</span>
          </div>
        </div>
      </div>
    </div>
  </section>

  <!-- PRÁTICA 4: CETOCONAZOL -->
  <section id="mod-p4" class="hidden space-y-6">
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b
border-slate-800 pb-4">
      <div>
        <span class="text-xs font-bold text-sky-400 uppercase tracking-widest">Prática 04 •
Espectrofotometria & Dissolução</span>
        <h2 class="text-2xl font-bold text-white">Análise de Absorção UV e Ion Trapping do
Cetoconazol</h2>
      </div>
      <button onclick="showModule('home')" class="px-3 py-1.5 rounded-lg bg-slate-800
hover:bg-slate-700 text-xs font-semibold text-slate-300">
        <i class="fa-solid fa-house mr-1"></i> Início
      </button>
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-12 gap-6">
      <!-- Interactive Spectrophotometer -->
      <div class="lg:col-span-12 xl:col-span-7 glass-card p-6 rounded-xl space-y-6">
        <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center
space-x-2">
          <i class="fa-solid fa-wave-square text-sky-400"></i>
          <span>Simulador Espectrofotométrico UV (223 nm e 269 nm)</span>
        </h3>

        <div class="grid grid-cols-1 md:grid-cols-2 gap-6">
          <!-- Spectro Screen -->
          <div class="bg-slate-900/90 p-4 rounded-xl border border-slate-800 flex flex-col
justify-between space-y-4">
            <div class="flex justify-between items-center">
              <span class="text-[10px] font-mono text-sky-400">UV-VIS
SPECTROPHOTOMETER</span>
              <span class="w-2 h-2 rounded-full bg-emerald-500 animate-pulse"></span>
            </div>

            <!-- Cuvette Holder graphic -->
            <div class="h-32 bg-slate-950 rounded-lg border border-slate-800 flex items-center
justify-center relative overflow-hidden">
```

```
            <div id="uv-beam" class="absolute w-full h-1 bg-cyan-400/80 blur-xs
transition-opacity duration-300 opacity-80"></div>
            <!-- Cuvette -->
            <div class="w-12 h-20 bg-slate-200/20 border-2 border-slate-300/40 rounded-sm
flex items-end p-1 z-10 relative">
              <div id="cuvette-liquid" class="w-full bg-sky-500/30 rounded-xs transition-all
duration-500" style="height: 80%;"></div>
            </div>
          </div>

          <div class="bg-slate-950 p-3 rounded border border-slate-800 flex justify-between
items-center">
            <span class="text-xs text-slate-400">Absorbância ($\lambda = 223 nm$):</span>
            <span id="p4-abs-val" class="text-lg font-bold font-mono text-cyan-400">0.850
AU</span>
          </div>
        </div>

        <!-- Interactivity Controls -->
        <div class="space-y-4">
          <div class="bg-slate-800/60 p-4 rounded-xl border border-slate-700 space-y-3">
            <label class="text-xs font-bold text-slate-300 block">Simular Co-administração
com Antiácidos:</label>
            <div class="space-y-2">
              <button onclick="setP4Environment('acid')" id="p4-btn-acid" class="w-full p-2.5
rounded-lg border text-left text-xs font-bold bg-sky-500/20 border-sky-500 text-white flex
justify-between items-center">
                <span>Apenas Suco Gástrico (pH 1.2)</span>
                <i class="fa-solid fa-check text-sky-400"></i>
              </button>
              <button onclick="setP4Environment('antacid')" id="p4-btn-antacid" class="w-full
p-2.5 rounded-lg border border-slate-700 bg-slate-800/50 text-slate-400
hover:bg-slate-700/50 text-xs font-bold flex justify-between items-center">
                <span>Com Antiácido / Omeprazol (pH 6.0)</span>
                <i class="fa-solid fa-ban text-slate-600"></i>
              </button>
            </div>
          </div>

          <div id="p4-status-box" class="p-3 bg-slate-800 rounded-lg border border-slate-700
text-xs text-slate-300">
            <span class="font-bold text-emerald-400 block mb-1">Dissolução
Completa!</span>
            <p class="text-slate-400 text-[11px]">
              O Cetoconazol é uma base fraca dibásica. Em meio fortemente ácido ($pH <
2$), protona-se completamente, permitindo dissolução total e absorção ideal.
            </p>
          </div>
```

```
          </div>
        </div>
      </div>

      <!-- Ion Trapping Concept -->
      <div class="lg:col-span-12 xl:col-span-5 glass-card p-6 rounded-xl space-y-4">
        <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center space-x-2">
          <i class="fa-solid fa-baby text-sky-400"></i>
          <span>Conceito de Aprisionamento Iônico (Ion Trapping)</span>
        </h3>

        <div class="space-y-3 text-xs leading-relaxed text-slate-300">
          <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700">
            <h4 class="font-bold text-sky-300 mb-1">Passagem para o Leite Materno:</h4>
            <p class="text-slate-400">
              O plasma possui pH $\approx 7.4$, enquanto o leite materno é ligeiramente mais ácido (pH $\approx 6.8 - 7.0$).
            </p>
            <p class="text-slate-400 mt-2">
              Bases fracas como o Cetoconazol cruzam a barreira mamária na sua forma lipofílica não-ionizada. Ao entrarem no leite materno mais ácido, protonam-se ($BH^+$), tornando-se ionizadas e incapazes de retornar ao plasma.
            </p>
          </div>
        </div>
      </div>
    </div>
  </section>

  <!-- PRÁTICA 5: SULFONAMIDAS -->
  <section id="mod-p5" class="hidden space-y-6">
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-800 pb-4">
      <div>
        <span class="text-xs font-bold text-emerald-400 uppercase tracking-widest">Prática 05 • QSAR & Relação Estrutura-Atividade</span>
        <h2 class="text-2xl font-bold text-white">CCD em Fase Reversa e Parâmetros Hidrofóbicos de Sulfonamidas</h2>
      </div>
      <button onclick="showModule('home')" class="px-3 py-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-xs font-semibold text-slate-300">
        <i class="fa-solid fa-house mr-1"></i> Início
      </button>
    </div>

    <div class="grid grid-cols-1 lg:grid-cols-12 gap-6">
      <!-- QSAR Table & Data -->
```

```
<div class="lg:col-span-12 xl:col-span-7 glass-card p-6 rounded-xl space-y-4">
  <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center space-x-2">
    <i class="fa-solid fa-table text-emerald-400"></i>
    <span>Série Homóloga de Sulfonamidas - Parâmetros de Hansch ($\pi$)</span>
  </h3>

  <div class="overflow-x-auto">
    <table class="w-full text-left text-xs border-collapse">
      <thead>
        <tr class="border-b border-slate-700 text-slate-400 font-mono">
          <th class="p-2">Análogo / Substituinte (R)</th>
          <th class="p-2">$\pi$ (Hansch)</th>
          <th class="p-2">$R_f$ (Fase Reversa C18)</th>
          <th class="p-2">$R_m = \log(1/R_f - 1)$</th>
          <th class="p-2">MIC ($\mu M$)</th>
        </tr>
      </thead>
      <tbody class="divide-y divide-slate-800 text-slate-200">
        <tr class="hover:bg-slate-800/40">
          <td class="p-2 font-bold text-emerald-400">Sulfanilamida (-H)</td>
          <td class="p-2 font-mono">0.00</td>
          <td class="p-2 font-mono">0.82</td>
          <td class="p-2 font-mono">-0.66</td>
          <td class="p-2 font-mono">120.0</td>
        </tr>
        <tr class="hover:bg-slate-800/40">
          <td class="p-2 font-bold text-emerald-400">Sulfametazina (-CH3)</td>
          <td class="p-2 font-mono">0.56</td>
          <td class="p-2 font-mono">0.60</td>
          <td class="p-2 font-mono">-0.18</td>
          <td class="p-2 font-mono">25.0</td>
        </tr>
        <tr class="hover:bg-slate-800/40">
          <td class="p-2 font-bold text-emerald-400">Sulfametoxazol (-OMe)</td>
          <td class="p-2 font-mono">0.98</td>
          <td class="p-2 font-mono">0.42</td>
          <td class="p-2 font-mono">0.14</td>
          <td class="p-2 font-mono">4.5 (Ótimo)</td>
        </tr>
        <tr class="hover:bg-slate-800/40">
          <td class="p-2 font-bold text-emerald-400">Sulfadiazina Bromo (-Br)</td>
          <td class="p-2 font-mono">1.86</td>
          <td class="p-2 font-mono">0.20</td>
          <td class="p-2 font-mono">0.60</td>
          <td class="p-2 font-mono">85.0</td>
        </tr>
      </tbody>
```

```
          </table>
        </div>
      </div>

      <!-- Parabolic QSAR Chart -->
      <div class="lg:col-span-12 xl:col-span-5 glass-card p-6 rounded-xl space-y-4">
        <h3 class="text-sm font-bold text-slate-300 uppercase tracking-wider flex items-center space-x-2">
          <i class="fa-solid fa-chart-area text-emerald-400"></i>
          <span>Curva Parabólica de Hansch ($\log 1/C$ vs $\pi$)</span>
        </h3>

        <div class="h-56 bg-slate-900/80 p-2 rounded-xl border border-slate-800">
          <canvas id="qsarChart"></canvas>
        </div>

        <div class="p-3 bg-slate-800/60 rounded-lg border border-slate-700 text-xs text-slate-300">
          <span class="font-bold text-emerald-400 block mb-1">Lipofilicidade Ótima ($\log P_0$):</span>
          <p class="text-slate-400 text-[11px]">
            Aumentar a lipofilicidade melhora a permeabilidade na membrana bacteriana até atingir $\log P_0$. Além desse ponto, o fármaco fica retido nas membranas lipídicas ou precipita no meio aquoso.
          </p>
        </div>
      </div>
    </div>
  </section>

  <!-- QUIZ / GAMIFICATION MODULE -->
  <section id="mod-quiz" class="hidden space-y-6">
    <div class="flex flex-col md:flex-row md:items-center justify-between gap-4 border-b border-slate-800 pb-4">
      <div>
        <span class="text-xs font-bold text-amber-400 uppercase tracking-widest">Avaliação & Gamificação</span>
        <h2 class="text-2xl font-bold text-white">Quiz Prático QMFXperience</h2>
      </div>
      <button onclick="showModule('home')" class="px-3 py-1.5 rounded-lg bg-slate-800 hover:bg-slate-700 text-xs font-semibold text-slate-300">
        <i class="fa-solid fa-house mr-1"></i> Início
      </button>
    </div>

    <div class="max-w-3xl mx-auto glass-card p-6 lg:p-8 rounded-2xl space-y-6 border border-amber-500/20">
      <!-- Progress Bar -->
```

```
        <div class="flex justify-between items-center text-xs text-slate-400 mb-2">
          <span id="quiz-progress-text">Questão 1 de 5</span>
          <span id="quiz-score-text" class="font-bold text-amber-400">Pontuação: 0</span>
        </div>
        <div class="w-full bg-slate-800 rounded-full h-2">
          <div id="quiz-progress-bar" class="bg-gradient-to-r from-amber-500 to-orange-500 h-2
rounded-full transition-all duration-300" style="width: 20%;"></div>
        </div>

        <!-- Question Container -->
        <div id="quiz-container" class="space-y-6">
          <h3 id="quiz-question" class="text-lg font-bold text-white leading-relaxed">
            Pergunta...
          </h3>

          <!-- Options -->
          <div id="quiz-options" class="space-y-3">
            <!-- Dynamic Options -->
          </div>
        </div>

        <!-- Feedback & Next Button -->
        <div id="quiz-feedback" class="hidden p-4 rounded-xl text-xs font-medium
space-y-2"></div>

        <div class="flex justify-end pt-4 border-t border-slate-800">
          <button id="quiz-next-btn" onclick="nextQuestion()" class="hidden px-5 py-2.5
rounded-xl bg-gradient-to-r from-amber-500 to-orange-600 text-white font-bold text-xs
hover:from-amber-600 hover:to-orange-700 transition-all shadow-md">
            Próxima Questão <i class="fa-solid fa-arrow-right ml-1"></i>
          </button>
        </div>
      </div>
    </section>

  </main>

  <!-- FOOTER -->
  <footer class="glass-nav py-4 px-8 text-center text-xs text-slate-500 mt-12 border-t
border-slate-800">
    <p>QMFXperience • Desenvolvido para o Departamento de Farmácia da Universidade
Federal do Paraná (UFPR)</p>
  </footer>

  <!-- JAVASCRIPT LOGIC -->
  <script>
    // NAVIGATION CONTROLLER
    function showModule(modId) {
```

```
    const modules = ['home', 'p1', 'p2', 'p3', 'p4', 'p5', 'quiz'];
    modules.forEach(id => {
      const el = document.getElementById(`mod-${id}`);
      if(el) el.classList.add('hidden');

      const navEl = document.getElementById(`nav-${id}`);
      if(navEl) {
        navEl.classList.remove('bg-slate-700', 'text-white');
        navEl.classList.add('text-slate-300');
      }
    });

    const activeMod = document.getElementById(`mod-${modId}`);
    if(activeMod) activeMod.classList.remove('hidden');

    const activeNav = document.getElementById(`nav-${modId}`);
    if(activeNav) {
      activeNav.classList.add('bg-slate-700', 'text-white');
    }

    window.scrollTo({ top: 0, behavior: 'smooth' });

    // Init charts if needed
    if(modId === 'p3' && !captoprilChart) initP3Chart();
    if(modId === 'p5' && !qsarChart) initQSARChart();
    if(modId === 'quiz') startQuiz();
  }

  // PRÁTICA 1 LOGIC
  function setP1Step(step) {
    [1,2,3].forEach(s => {
      const btn = document.getElementById(`p1-btn-${s}`);
      if(btn) {
        btn.className = 'p-3 rounded-lg border border-slate-700 bg-slate-800/50
text-slate-400 hover:bg-slate-700/50';
      }
    });

    const activeBtn = document.getElementById(`p1-btn-${step}`);
    if(activeBtn) {
      activeBtn.className = 'p-3 rounded-lg border text-left transition-all bg-cyan-500/20
border-cyan-500 text-white';
    }

    const v1 = document.getElementById('p1-view-1');
    const v2 = document.getElementById('p1-view-2');

    if(step === 1) {
```

```
    v1.classList.remove('hidden');
    v2.classList.add('hidden');
  } else {
    v1.classList.add('hidden');
    v2.classList.remove('hidden');
  }
}

// PRÁTICA 2 LOGIC
function updateExtraction() {
  const vol = document.getElementById('vol-eter').value;
  document.getElementById('vol-eter-val').innerText = `${vol} mL`;

  const ph = parseFloat(document.getElementById('p2-ph').value);

  let cOrg = 0.024;
  let cAq = 0.008;

  if(ph === 3.8) {
    cOrg = 0.012;
    cAq = 0.020;
  } else if(ph === 6.0) {
    cOrg = 0.001;
    cAq = 0.038;
  }

  document.getElementById('c-org-val').innerText = `${cOrg.toFixed(3)} M`;
  document.getElementById('c-aq-val').innerText = `${cAq.toFixed(3)} M`;

  const logP = Math.log10(cOrg / cAq);
  document.getElementById('logp-val').innerText = logP.toFixed(2);
}

function runExtraction() {
  const org = document.getElementById('phase-org');
  const aq = document.getElementById('phase-aq');
  org.classList.add('opacity-50');
  aq.classList.add('opacity-50');
  setTimeout(() => {
    org.classList.remove('opacity-50');
    aq.classList.remove('opacity-50');
  }, 500);
}

// PRÁTICA 3 LOGIC (CHART.JS)
let captoprilChart = null;
const p3DataRaw = [
  { v: 0, ph: 2.1 }, { v: 2, ph: 2.5 }, { v: 4, ph: 3.1 }, { v: 6.25, ph: 3.7 },
```

```
  { v: 8, ph: 4.5 }, { v: 10, ph: 6.0 }, { v: 12.5, ph: 9.8 }, { v: 15, ph: 11.5 }
];

function initP3Chart() {
  const ctx = document.getElementById('captoprilChart').getContext('2d');
  captoprilChart = new Chart(ctx, {
    type: 'line',
    data: {
      labels: p3DataRaw.map(d => d.v + ' mL'),
      datasets: [{
        label: 'pH do Meio',
        data: p3DataRaw.map(d => d.ph),
        borderColor: '#818cf8',
        backgroundColor: 'rgba(129, 140, 248, 0.1)',
        fill: true,
        tension: 0.4
      }]
    },
    options: {
      responsive: true,
      maintainAspectRatio: false,
      plugins: { legend: { display: false } },
      scales: {
        x: { grid: { color: '#334155' }, ticks: { color: '#94a3b8', font: { size: 10 } } },
        y: { grid: { color: '#334155' }, ticks: { color: '#94a3b8', font: { size: 10 } } }
      }
    }
  });
}

function updateP3Chart(type) {
  if(!captoprilChart) return;

  if(type === 'raw') {
    captoprilChart.data.datasets[0].data = p3DataRaw.map(d => d.ph);
    captoprilChart.data.datasets[0].borderColor = '#818cf8';
  } else if(type === 'd1') {
    captoprilChart.data.datasets[0].data = [0.2, 0.3, 0.8, 1.8, 0.6, 1.2, 3.5, 0.4];
    captoprilChart.data.datasets[0].borderColor = '#38bdf8';
  } else {
    captoprilChart.data.datasets[0].data = [0, 0.1, 0.5, 0, -0.6, 1.0, 0, -2.0];
    captoprilChart.data.datasets[0].borderColor = '#34d399';
  }
  captoprilChart.update();
}

function calcIonization() {
  const ph = parseFloat(document.getElementById('p3-ph-slider').value);
```

```
    document.getElementById('p3-ph-val').innerText = ph.toFixed(1);

    // pKa1 = 3.7 (carboxyl)
    const alpha1 = (1 / (1 + Math.pow(10, 3.7 - ph))) * 100;
    // pKa2 = 9.8 (thiol)
    const alpha2 = (1 / (1 + Math.pow(10, 9.8 - ph))) * 100;

    document.getElementById('p3-carboxyl-val').innerText = alpha1.toFixed(2) + '%';
    document.getElementById('p3-carboxyl-bar').style.width = alpha1 + '%';

    document.getElementById('p3-thiol-val').innerText = alpha2.toFixed(2) + '%';
    document.getElementById('p3-thiol-bar').style.width = alpha2 + '%';

    const site = document.getElementById('p3-site-val');
    if(ph < 3.0) {
      site.innerText = "Estômago (pH 1.5 - 2.0) - Predomina Não-Ionizado (Boa Absorção)";
      site.className = "block font-semibold text-emerald-400 mt-0.5";
    } else {
      site.innerText = "Intestino / Plasma - Ionizado (Alta Solubilidade / Absorção Ativa)";
      site.className = "block font-semibold text-indigo-400 mt-0.5";
    }
  }

  // PRÁTICA 4 LOGIC
  function setP4Environment(env) {
    const btnAcid = document.getElementById('p4-btn-acid');
    const btnAntacid = document.getElementById('p4-btn-antacid');
    const absVal = document.getElementById('p4-abs-val');
    const liquid = document.getElementById('cuvette-liquid');
    const statusBox = document.getElementById('p4-status-box');

    if(env === 'acid') {
      btnAcid.className = 'w-full p-2.5 rounded-lg border text-left text-xs font-bold
bg-sky-500/20 border-sky-500 text-white flex justify-between items-center';
      btnAntacid.className = 'w-full p-2.5 rounded-lg border border-slate-700
bg-slate-800/50 text-slate-400 hover:bg-slate-700/50 text-xs font-bold flex justify-between
items-center';
      absVal.innerText = '0.850 AU';
      liquid.className = 'w-full bg-sky-500/30 rounded-xs transition-all duration-500';
      statusBox.innerHTML = `
        <span class="font-bold text-emerald-400 block mb-1">Dissolução Completa!</span>
        <p class="text-slate-400 text-[11px]">
          O Cetoconazol protona-se completamente em pH ácido, garantindo dissolução e alta
absorção sistêmica.
        </p>
      `;
    } else {
```

```
      btnAntacid.className = 'w-full p-2.5 rounded-lg border text-left text-xs font-bold bg-amber-500/20 border-amber-500 text-white flex justify-between items-center';
      btnAcid.className = 'w-full p-2.5 rounded-lg border border-slate-700 bg-slate-800/50 text-slate-400 hover:bg-slate-700/50 text-xs font-bold flex justify-between items-center';
      absVal.innerText = '0.045 AU';
      liquid.className = 'w-full bg-slate-700/80 rounded-xs transition-all duration-500';
      statusBox.innerHTML = `
        <span class="font-bold text-amber-400 block mb-1">Falha de Dissolução / Precipitação!</span>
        <p class="text-slate-400 text-[11px]">
          Antiácidos elevam o pH gástrico. A base fraca permanece na sua forma neutra insolúvel, resultando em drástica redução na biodisponibilidade.
        </p>
      `;
    }
  }

  // PRÁTICA 5 LOGIC (QSAR CHART)
  let qsarChart = null;
  function initQSARChart() {
    const ctx = document.getElementById('qsarChart').getContext('2d');
    qsarChart = new Chart(ctx, {
      type: 'line',
      data: {
        labels: ['0.0', '0.5', '1.0 (Opt)', '1.5', '2.0'],
        datasets: [{
          label: 'Atividade Antibacteriana (log 1/C)',
          data: [0.8, 1.6, 2.5, 1.7, 0.6],
          borderColor: '#10b981',
          backgroundColor: 'rgba(16, 185, 129, 0.1)',
          fill: true,
          tension: 0.4
        }]
      },
      options: {
        responsive: true,
        maintainAspectRatio: false,
        plugins: { legend: { display: false } },
        scales: {
          x: { grid: { color: '#334155' }, ticks: { color: '#94a3b8', font: { size: 10 } } },
          y: { grid: { color: '#334155' }, ticks: { color: '#94a3b8', font: { size: 10 } } }
        }
      }
    });
  }

  // QUIZ SYSTEM LOGIC
  const quizQuestions = [
```

```
    {
      q: "Qual é o objetivo da hidrólise alcalina na Sinvastatina durante o processo de bioativação?",
      options: [
        "Converter o carboxilato na forma de éster apolar",
        "Abrir o anel lactônico expondo o hidroxiácido ativo",
        "Aumentar a lipofilicidade do fármaco para cruzar a BHE",
        "Inativar a molécula antes da eliminação renal"
      ],
      correct: 1,
      explanation: "A Sinvastatina é um pró-fármaco lactônico. A hidrólise abre o anel para formar o hidroxiácido ativo com grupo carboxilato ionizado."
    },
    {
      q: "Na análise de CCD em Fase Normal (Sílica Gel), qual o comportamento do Ácido Sinvastatínico comparado à Sinvastatina?",
      options: [
        "O Ácido possui maior Rf pois é mais apolar",
        "A Sinvastatina possui menor Rf pois interage forte com a sílica",
        "O Ácido possui menor Rf devido à interações polares com a fase estacionária",
        "Ambos eluem no mesmo Rf independentemente do eluente"
      ],
      correct: 2,
      explanation: "Na fase normal, a sílica é polar. O ácido ativado é mais polar e fica mais retido (menor Rf)."
    },
    {
      q: "Por que a co-administração de Omeprazol reduz drasticamente a absorção do Cetoconazol?",
      options: [
        "Aumenta a degradação enzimática no fígado",
        "Eleva o pH gástrico, impedindo a protonação e dissolução da base fraca",
        "Inibe os transportadores ativos no intestino",
        "Acelera o esvaziamento gástrico"
      ],
      correct: 1,
      explanation: "O Cetoconazol precisa de pH gástrico bastante ácido para se protonar e se dissolver adequadamente."
    }
  ];

  let currentQ = 0;
  let score = 0;

  function startQuiz() {
    currentQ = 0;
    score = 0;
    renderQuestion();
```

```
  }

  function renderQuestion() {
    const q = quizQuestions[currentQ];
    document.getElementById('quiz-progress-text').innerText = `Questão ${currentQ + 1} de
${quizQuestions.length}`;
    document.getElementById('quiz-score-text').innerText = `Pontuação: ${score}`;
    document.getElementById('quiz-progress-bar').style.width = `${((currentQ +
1)/quizQuestions.length)*100}%`;

    document.getElementById('quiz-question').innerText = q.q;

    const optsContainer = document.getElementById('quiz-options');
    optsContainer.innerHTML = '';

    q.options.forEach((opt, idx) => {
      const btn = document.createElement('button');
      btn.className = 'w-full p-3.5 rounded-xl border border-slate-700 bg-slate-800/60
hover:bg-slate-700/60 text-slate-200 font-medium text-xs text-left transition-all flex
items-center justify-between group';
      btn.onclick = () => checkAnswer(idx);
      btn.innerHTML = `
        <span>${opt}</span>
        <i class="fa-regular font-bold text-slate-500 group-hover:text-amber-400"></i>
      `;
      optsContainer.appendChild(btn);
    });

    document.getElementById('quiz-feedback').className = 'hidden';
    document.getElementById('quiz-next-btn').className = 'hidden';
  }

  function checkAnswer(selectedIdx) {
    const q = quizQuestions[currentQ];
    const feedback = document.getElementById('quiz-feedback');
    feedback.classList.remove('hidden');

    if(selectedIdx === q.correct) {
      score += 100;
      feedback.className = 'p-4 rounded-xl text-xs font-medium space-y-1
bg-emerald-500/10 border border-emerald-500/30 text-emerald-300';
      feedback.innerHTML = `<span class="font-bold flex items-center"><i class="fa-solid
fa-circle-check mr-2"></i> Resposta Correta!</span><p class="text-[11px]
text-slate-300">${q.explanation}</p>`;
    } else {
      feedback.className = 'p-4 rounded-xl text-xs font-medium space-y-1 bg-rose-500/10
border border-rose-500/30 text-rose-300';
```

```
      feedback.innerHTML = `<span class="font-bold flex items-center"><i class="fa-solid fa-circle-xmark mr-2"></i> Resposta Incorreta</span><p class="text-[11px] text-slate-300">${q.explanation}</p>`;
    }

    document.getElementById('quiz-score-text').innerText = `Pontuação: ${score}`;
    document.getElementById('quiz-next-btn').className = 'px-5 py-2.5 rounded-xl bg-gradient-to-r from-amber-500 to-orange-600 text-white font-bold text-xs hover:from-amber-600 hover:to-orange-700 transition-all shadow-md';
  }

  function nextQuestion() {
    currentQ++;
    if(currentQ < quizQuestions.length) {
      renderQuestion();
    } else {
      showQuizResults();
    }
  }

  function showQuizResults() {
    document.getElementById('quiz-container').innerHTML = `
      <div class="text-center py-8 space-y-4">
        <div class="w-16 h-16 rounded-full bg-amber-500/20 text-amber-400 flex items-center justify-center text-3xl mx-auto border border-amber-500/30">
          <i class="fa-solid fa-trophy"></i>
        </div>
        <h3 class="text-2xl font-bold text-white">Parabéns! Desafio Concluído!</h3>
        <p class="text-slate-300 text-xs">Você obteve <span class="font-bold text-amber-400 font-mono">${score} pontos</span> no simulador da feira de QMF-UFPR.</p>
        <button onclick="startQuiz()" class="px-5 py-2.5 rounded-xl bg-slate-800 hover:bg-slate-700 text-white font-bold text-xs transition-all">
          Refazer Quiz
        </button>
      </div>
    `;
    document.getElementById('quiz-feedback').className = 'hidden';
    document.getElementById('quiz-next-btn').className = 'hidden';
  }

  // Default View Initialization
  showModule('home');
 </script>
</body>
</html>
```